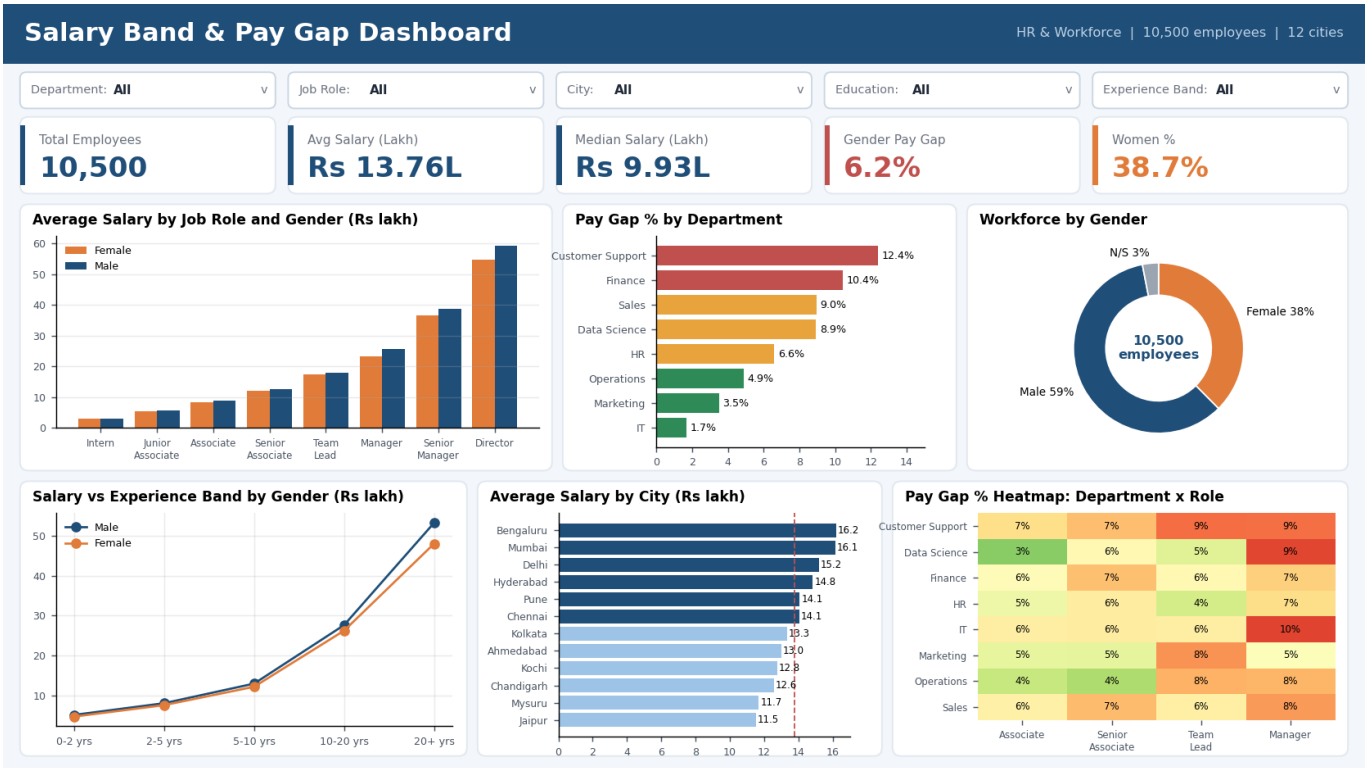
DA-23 Salary Band & Pay Gap Analysis

Step 5: Power BI Dashboard Build Guide

Final dashboard preview (what your Power BI page will look like):



Design idea: one 16:9 page. Top: five dropdown filters. Then five KPI cards. Then six visuals that answer the business question "Is pay consistent across roles and gender?". Colours are consistent everywhere: **orange = Female, navy = Male**, and the pay gap uses green / amber / red.

Files you need

File	Purpose
employee_powerbi_data.csv	Cleaned data (10,500 rows, 18 columns) with extra columns Salary_Lakh, Experience_Band, Joining_Year
DA23_PowerBI_Theme.json	Colour theme, fonts, rounded white cards, light grey page background
DA23_DAX_Measures.txt	All measures and helper columns, ready to copy and paste

Step-by-step build (about 45 minutes)

Step 1: Load the data

Open Power BI Desktop → **Home** → **Get data** → **Text/CSV** → choose *employee_powerbi_data.csv* → **Transform Data**. Check types: Annual_Salary_INR, Age, Job_Level, Performance_Rating = Whole number; Experience_Years, Salary_Lakh = Decimal; Joining_Date = Date. Rename the query to **Employees** → **Close & Apply**.

Step 2: Apply the theme

View → **Themes** → **Browse for themes** → select *DA23_PowerBI_Theme.json*. This sets the colours, fonts, rounded corners and the light page background.

Step 3: Page setup

Click the empty page → **Format** → **Canvas settings** → **Type: 16:9**. Turn on **View** → **Gridlines** and **Snap to grid**. Rename the page tab to "Pay Gap Dashboard".

Step 4: Add the DAX measures

Right-click the Employees table → **New measure**. Paste each measure from *DA23_DAX_Measures.txt* one by one (list on page 4). Also create the two helper columns **Band Sort** and **Exp Sort**.

Step 5: Fix the sort orders

Select the column **Job_Role** → Column tools → **Sort by column** → **Job_Level**. Do the same for **Experience_Band** → **Exp Sort** and **Salary_Band** → **Band Sort**. Without this, roles and bands appear alphabetically.

Step 6: Build the header

Insert → **Shapes** → **Rectangle**: size 1280 x 56, position 0,0, fill #1F4E78, no border. **Insert** → **Text box** on top: "Salary Band & Pay Gap Dashboard", Segoe UI Semibold, 20 pt, white. Optional: a Card with **Gap Headline** on the right in white text.

Step 7: Add slicers, KPI cards and visuals

Follow the specification table on page 3. Set each visual's size and position in **Format** → **General** → **Properties** so everything lines up exactly.

Step 8: Polish and save

See the polish checklist on page 5. **File** → **Save as** DA23_Pay_Gap_Dashboard.pbix. Optional: **Publish** to Power BI Service for a shareable link.

No Windows PC? Power BI Desktop runs only on Windows. Use the Power BI Service web app (app.powerbi.com) to upload the CSV and build the same visuals, though theme import and sort-by-column options are limited there.

Visual specification (position in pixels on a 1280 x 720 canvas)

Element	Visual type	Fields and settings	x, y, w, h
Slicers (5)	Slicer → **Dropdown** style	Department, Job_Role, City, Education, Experience_Band	y=64, h=34, w=240; x = 16, 268, 520, 772, 1024
KPI card 1	Card	[Total Employees]	16,106,240,72
KPI card 2	Card	[Avg Salary (Lakh)] format: Rs 0.00 "L"	268,106,240,72
KPI card 3	Card	[Median Salary (Lakh)]	520,106,240,72
KPI card 4	Card	[Gender Pay Gap %] font colour: fx → Field value → [Gap Colour]	772,106,240,72
KPI card 5	Card	[Women %]	1024,106,240,72
V1 Salary by role and gender	Clustered column chart	X: Job_Role | Y: [Avg Salary (Lakh)] | Legend: Gender | visual filter: Gender is Female or Male	16,188,500,250
V2 Pay gap by department	Clustered bar chart	Y: Department | X: [Gender Pay Gap %] | data labels on | sort high to low | bar colour: fx → [Gap Colour] | filter: Department is not Unknown	526,188,370,250
V3 Workforce by gender	Donut chart	Legend: Gender | Values: [Total Employees] | detail labels: category and percent	906,188,358,250
V4 Salary vs experience	Line chart with markers	X: Experience_Band | Y: [Avg Salary (Lakh)] | Legend: Gender | filter Gender is Female or Male	16,448,420,258
V5 Salary by city	Clustered bar chart	Y: City | X: [Avg Salary (Lakh)] | sort high to low | Analytics → Average line (red dashed) | filter City is not Unknown	446,448,380,258
V6 Pay gap heatmap	Matrix	Rows: Department | Columns: Job_Role | Values: [Gender Pay Gap %] | Cell elements → Background colour → Gradient (green low, yellow mid, red high)	836,448,428,258

Important: Do not add a Gender slicer. The pay gap measures compare Male and Female on purpose, so filtering by gender would break them. If you want a gender view, use the donut chart.

Every KPI card and chart should have a clear title and units (Rs lakh, %). Turn off the visual header icons (the theme already does this) so the page looks clean.

DAX measures (copy from DA23_DAX_Measures.txt)

```
Total Employees = COUNTROWS ( Employees )

Avg Salary = AVERAGE ( Employees[Annual_Salary_INR] )

Avg Salary (Lakh) = DIVIDE ( [Avg Salary], 100000 )

Median Salary (Lakh) = DIVIDE ( MEDIAN ( Employees[Annual_Salary_INR] ), 100000 )

Male Avg Salary = CALCULATE ( [Avg Salary], Employees[Gender] = "Male" )

Female Avg Salary = CALCULATE ( [Avg Salary], Employees[Gender] = "Female" )

Gender Pay Gap % =
DIVIDE ( [Male Avg Salary] - [Female Avg Salary], [Male Avg Salary] )

Women % =
DIVIDE (
    CALCULATE ( COUNTROWS ( Employees ), Employees[Gender] = "Female" ),
    CALCULATE ( COUNTROWS ( Employees ), Employees[Gender] IN { "Male", "Female" } )
)

Gap Colour =
SWITCH ( TRUE (),
    [Gender Pay Gap %] >= 0.10, "#C0504D",
    [Gender Pay Gap %] >= 0.06, "#E8A33D",
    "#2E8B57" )

Gap Headline =
"Women earn " & FORMAT ( [Gender Pay Gap %], "0.0%" ) & " less than men in the current selection"

Exp Sort (column) =
SWITCH ( Employees[Experience_Band], "0-2 yrs", 1, "2-5 yrs", 2,
         "5-10 yrs", 3, "10-20 yrs", 4, "20+ yrs", 5 )
```

Format [Gender Pay Gap %] and [Women %] as **Percentage, 1 decimal**; format [Avg Salary (Lakh)] and [Median Salary (Lakh)] as **Decimal, 2 places**. Set the format in **Measure tools**. Expected card values with no filters: 10,500 employees, Rs 13.76 L average, Rs 9.93 L median, 6.2% gap, 38.7% women.

Quick test: after building, set the Department slicer to **IT**. The gap card should change to about 1.7%. Select **Customer Support** and it should show about 12.4%.

Make it more attractive: polish checklist

- **Align everything:** select all visuals → **Format** → **Align** and **Distribute horizontally**. Equal gaps (about 10 px) make it look professional.
- **Accent bars on KPI cards:** add a thin rectangle (5 x 56 px) on the left edge of each card, navy for most, red for the gap card, orange for Women %.
- **Keep colours consistent:** Female = orange #E07B39, Male = navy #1F4E78. If the legend colours swap, set them manually in Format → Visual → Columns → Colours.
- **Reset button:** View → Bookmarks → Add a bookmark with no filters → Insert → Buttons → Bookmark, labelled "Reset filters".
- **Tooltip page (optional):** a hidden page with a small card showing [Gap Headline], set as the tooltip for V2.
- **Page 2 "Insights" (optional):** a text page with the 4 recommendations from Step 6, for the presentation.
- **Fonts:** titles 12 pt semibold, labels 8 to 10 pt, KPI numbers 28 pt. Never use more than two font sizes for labels.
- **Remove clutter:** turn off gridlines in charts if busy, show data labels only where they help, hide axis titles that repeat the chart title.

Rubric check (Dashboard, 20 marks: clear layout, correct KPIs, useful filters, easy to read)

Criterion	How this dashboard meets it
Clear layout	Header, filters, KPIs, then charts, in a Z-pattern reading order; equal spacing
Correct KPIs	Employees, average and median salary, gender pay gap, women %; values match your SQL and Python results (6.24%, Rs 13.76 L, Rs 9.93 L)
Useful filters	Department, Job Role, City, Education, Experience Band dropdowns; all visuals respond
Easy to read	One consistent colour code, titles with units, 6 visuals only, no clutter
4-6 visuals	Column, bar, donut, line, bar, matrix heatmap (6)

What to say about it in your viva (5 minutes)

- Pay is **not consistent**: women earn **6.2% less** overall, and the gap appears in every job role (V1) and every experience band (V4).
- The heatmap (V6) shows the problem is widest in Customer Support, Finance and Manager level roles; IT is the closest to equal pay.
- City matters too: Bengaluru and Mumbai pay about 29% more than Jaipur (V5).
- Women are about 38% of staff and balanced across roles (V3), so the issue is how pay is set, not who is hired.
- Click a department slicer live to show how the KPIs and gap change.